Retail Sales & Customer Behaviour Analysis

1. Problem Statement

A Business Data Analyst at a UK-based retail firm is tasked with analysing transactional sales data to uncover insights related to revenue performance, customer behaviour, and product trends.

The primary objectives of this analysis are:

- Increase revenue
- Improve customer retention
- Optimise product strategy

2. Analytical Objectives

Sales Analysis

- What are the monthly revenue trends?
- Which products generate the most revenue?

Customer Analysis

- Who are the most valuable customers?
- What is the repeat purchase rate?

Product Insights

- Which products drive the highest revenue?
- Which products or categories have high return rates?

3. Dataset Overview

The dataset contains transactional-level retail data with the following fields:

- InvoiceNo: Unique identifier for each transaction
- StockCode: Product code
- Description: Product name/description
- Quantity: Number of units purchased
- InvoiceDate: Timestamp of the transaction
- UnitPrice: Price per unit

- CustomerID: Unique identifier for each customer
- Country: Country of purchase

4. Data Quality Assessment

Missing Values

- Missing values were identified in CustomerID and Description fields.

Returns Handling

- Negative values in the Quantity column represent product returns/refunds.

Time Coverage

- The dataset spans approximately one year (01/12/2010 – 09/12/2011), covering 373 days.

5. Data Preparation & Challenges

5.1 Datetime Formatting Issue

During data ingestion into SQL, the import process failed due to incompatible datetime formats.

- The dataset used UK format: DD/MM/YYYY
- MySQL requires ISO format: YYYY-MM-DD

This was resolved using Python by parsing the datetime field with `dayfirst=True` and converting it into a standardized format.

5.2 Datetime Data Type Handling

While attempting to format dates using `strftime()`, the datetime column was converted into a string type.

This led to the realization that:

Datetime *formatting* (display) is different from datetime *storage* (data type).

To preserve analytical capability, the column was retained as a datetime object and only formatted when required for presentation.

5.3 Product Categorisation (Feature Engineering)

The dataset did not include a product category field, which limited higher-level analysis.

To address this:

- A Category feature was engineered in Python
- Keyword-based classification was applied to group products into categories such as:
    - Home Decor
    - Kitchen
    - Storage
    - Accessories
    - Lighting
    - Seasonal & Gifts

Initial categorisation resulted in over 60% of records falling under “Other”, indicating insufficient coverage.

The classification logic was iteratively refined, reducing “Other” to ~35%, improving analytical usefulness.

6. Key Insights from SQL Analysis

6.1 Revenue Performance & Seasonality

- Total revenue for the period was approximately £8.3 million
- Revenue shows a consistent upward trend from May to November, with a peak in November (£1.13M)

This pattern likely reflects seasonal demand driven by:

- Black Friday
- Pre-Christmas shopping
- Revenue in December 2011 (£342.5K) was significantly lower than December 2010 (£554.6K)

This may indicate:

- Incomplete data for December 2011
- Post-peak seasonal decline

6.2 Product Revenue Concentration

- The top-selling product (REGENCY CAKESTAND 3 TIER) contributes only 1.6% of total revenue

- The top 10 products contribute 8.18% of total revenue

This indicates:

A highly diversified product portfolio with low dependency on individual products

6.3 High-Value, Low-Volume Products

Some products generate significant revenue despite low sales volume.

Example:

- PICNIC BASKET WICKER 60 PIECES
  - Revenue: £39.6K
  - Units sold: 61

This highlights the presence of premium products driving revenue through pricing rather than volume.

6.4 Customer Behaviour & Retention

- Top 10 customers contribute 16.5% of total revenue
- Highest individual customer contribution: £279K (3.37%)

This indicates moderate customer concentration.

- 69.96% of customers are repeat buyers

This suggests:

Customer retention is a key driver of revenue

Recommendation:

- Implement loyalty programs
- Personalised offers
- Lifecycle marketing strategies

6.5 Geographic Revenue Distribution

- UK contributes 81.54% of total revenue
- European markets (Netherlands, Ireland, Germany, France) contribute 11.49%
- Australia contributes 1.65%

This indicates:

- Strong domestic dependency
- Untapped international growth potential

However, lower international revenue may be due to:

- Limited brand awareness
- Logistical constraints
- Local demand differences

Further market-specific analysis is required before expansion decisions.

7. Power BI Dashboard

A dashboard was developed to visualise key insights, including:

- Revenue trends over time
- Product performance
- Category-level analysis
- Customer behaviour
- Return rate analysis

Returns Definition

In this analysis, returns refer to product returns (negative quantities) and not financial returns such as ROI.

8. Key Takeaways

- Revenue is seasonal, peaking during key retail periods
- The business benefits from a diversified product portfolio
- Customer retention plays a critical role in revenue generation
- Certain categories exhibit higher return rates, indicating potential operational issues
- There is significant opportunity for international expansion, subject to further analysis

9. Limitations

- No cost data available → profitability analysis not possible
- Product categories were engineered, not sourced from original dataset
- Dataset limited to ~1 year → may not capture long-term trends

10. Conclusion

This analysis demonstrates how transactional retail data can be transformed into actionable insights through:

- Data cleaning and preprocessing
- Feature engineering
- SQL-based analysis
- Data visualisation

The project highlights the importance of combining technical skills with business understanding to drive meaningful insights.